

Mitch Cochran



Classes

- There will be a live workshop
- The live workshop will be Monday – 09:30 to 12:15
- Generally, the notes will be available on the weekend prior to the class. The expectation is that students will read through the slides and then listen to the pre-recorded lecture.
- I am a fan of TED talks that give some outside perspective. They are short targeted lectures by experts in the field. Please watch those prior to class. Generally, they are for understanding and will not be tested on (unless stated).



Who am I....

- Mitch Cochran
 - Director for Information Technology – City of San Bernardino – 5 years
 - IS Manager – City of Monrovia – 20 years
 - OC Courts – Senior Technical Systems Specialist
 - IBM Advisory Systems Engineer
 - Private Industry – Corporate Product Manager
- Military
 - California State Guard – Part of National Guard
 - LT Colonel – C4 (in charge of logistics)
- CISM / CGCIO
- Please use my mcochran.it@gmail.com account for emails / questions



Goals

- Introduction to analytical decision making – so what is it...
- Statistics – the basis for making the determinations
- Exploring data – Do you know what you really have?
- AI
- Tools:
 - Excel as an analysis tool
 - SQL
 - Report writing
 - Power BI
 - Google Analytics
- What would you like to see covered?
- Content may change slightly due to class requests



Course Outline

Week 1	Introduction to Data Mining What is data? Overview of all Methods, Errors, Statistical Methods	Students choose when to present the topic article
Week 2	Exploratory Statistics, Exploratory Data Analysis	Discussion Questions
Week 3	Data Structures, Excel as an Analytical tool	Discussion Questions Excel Exercise 1 & 2
Week 4	Product Quality, KPI, Introduction to Access / SQL	Discussion Questions SQL Exercises
Week 5	Database Report Writing, PII, Knowledge Management	Discussion Questions SQL Reports
Week 6	Business Analysis	Discussion Questions
Week 7	Visualization Dashboards, Power BI	Discussion Questions Team Analysis 1
Week 8	Decision Theory, 5 Types of Analytics	Discussion Questions Power BI Exercise
Week 9	Social Media / Google Analytics	Discussion Questions
Week 10	Knowledge Representation / Expert Systems / Machine learning	Discussion Questions Google Analytics Due
Week 11	AI, Neural Net, Text Mining	Discussion Questions Team Analysis 2
Week 12	AI, ChatGPT, Generative AI	Discussion Questions AI Exercise
Week 13	Analytic trends, deep fakes, Prompt Engineering	Discussion Questions
Week 14	Presentations	Discussion Questions Team Analysis 3 Presentations
Week 15	Presentations	Team Analysis 3 Presentations All late assignments due by Friday night Final Online Exercise



So what will be expected of you?

- Individual Blackboard Discussion Questions - total 10 points
 - Data Analytics Presentation Real World Example - total 5 points
 - Exercises: - total 35 points
 - Excel-1, Excel-2, Access/SQL, Reports Exercise
 - Power Bi, Google Analytics, AI Exercise
 - Team Analysis 1 - total 10 points
 - Data Analytics Presentation – Advanced Techniques - total 5 points
 - Team Analysis 2 - total 10 points
 - Team Analysis 3 – with formal Presentation - total 10 points
 - Final Examination - total 15 points
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- All written assignments will be in business memo form
 - To, From, Date, Subject, in a BLUFF style



Individual presentations

- There are two presentations: one week 7 or before and one after
 - 1st Half– Analyze a real word example
 - 2nd Half– Describe an advanced technique
- The student finds an article on any topic related to the data driven decision making from any source
- It is a 2 (or so) page writeup with the 1st half discussing the issue of interest and the 2nd half discusses why you care as an analyst. It is in business memo form.
 - The hard copy is submitted to canvas
- You need to find 2 additional sources
 - If you regurgitate one source you are a parrot. I am interested in your opinion which you need a few different viewpoints to develop.
 - Cite the additional sources
- You will present it in class for 2-3 minutes (no PowerPoint) – what are the key concepts?
 - We will then expand on it in a class discussion

Team Analysis Projects

- Your boss will be giving you a business issue or opportunity. You will need to develop an analysis. The instructions will be brief and may be vague. You will have to determine what to do, how to do it.
 - Business goals, Issues/Analysis, Courses of Action
 - The more detailed the better the grade.
- Max team size is 2, You can be a team of 1
 - Project 1
 - Due Week 7
 - Project 2
 - Due Week 11
 - Project 3
 - Due Week 14
 - Will be formally presented in Week 14 and Week 15

Student Individual Exercises

- Excel Exercises
 - 2 Exercises– Intro to Data Mining
- Access Exercises
 - SQL Practice
 - Report Writing
- Power BI
 - Dashboard – Using Access
- Google Analytics
- AI - Exercise

When are things due?

- They will be due on the following week on Monday before class...
- So if it is assigned on week 2 – it will be due week 3 before the Monday class.
- Generally there is a 2 week grace period after the due date without assessment of a penalty,
 - After that assignments will not be accepted
- No assignments will be accepted after Friday night of week 15.



Grading Criteria

- As a university students, you are expected to solve business problems.
- In this class you will not be given a cookbook.
 - There are no dumb questions
 - Except – those that you can answer your self – then why are you asking????
 - But don't ask me just because you don't want to look it up yourself.
- You are expected to figure out a business solution to a problem and present it.

Grading Criteria

- Papers:
 - Content
 - Professional image
 - Your original material and analysis – little credit for other people's work
 - Citing other people's work.
- Presentations:
 - Do you understand what you are presenting?
 - Does it cover the topic completely?
 - Does it make sense to the audience?



Materials

- Assigned text – No assigned Text
- Software will be open source
- Except Excel / Access
 - I am not sure if open office has all of the features that we will be using – but I am not against trying it....
- Tests will be from class slides, provided materials and class discussions

AI Issues

So can AI do all of it?



Data Analytics

- Foundational Knowledge / Techniques

- Need a foundation of knowledge to be able to answer the question
- How do you apply any tool...
 - How do you explain and implement the results
- When is AI wrong, how do you know?

- AI

- Integrating new tools, Key tools include
 - AI assistants (ChatGPT, Gemini Adobe Firefly, Perplexity),
 - content creation (Canva, Copy.ai, Copilot),
 - automation (Zapier), and
 - meeting aids (Otter.ai).
- Develop an understanding of the human role and the AI role

Best AI Tools for Business

- 1. Adobe Firefly
- 2. ChatGPT Plus
- 3. Perplexity
- 4. Altered Studio
- 5. Midjourney
- 6. GitHub Copilot
- 7. Yoodli
- 8. Fliki
- 9. Notion AI
- 10. SlidesAI
- 11. Microsoft Copilot (Windows integration)
- 12. Jasper AI



Slide rules to Calculators

- We are transitioning to a new way of doing things....
 - Most of the time you don't know it until it is over, or it is too late.....
 - Lagging indicators
- My engineer father wanted me to learn a slide rule since he used it for 40 years. I wanted to use a calculator (\$100 to just add/subtract/multiply/divide)
 - But you still have to learn your multiplication techniques
 - You have to understand the context and foundations.
 - Calculators can look up info – Sine/Cosine/Tangent faster than we can, we don't have to calculate it but you know when the numbers are off.
- Now how do we use AI?



AI Research Notes

- We should not trust everything on the Internet - so why trust AI
 - Academic uses peer-reviewed for a reason
 - AI has hallucinations
- Do different AI systems give similar answers?
 - 3 different systems gave three different Latin translations for a motto
 - So which one do you believe?
 - Is there symmetry in that AI?
 - Will an AI response if reversed get the original question?
 - The Latin translations did not come back to the original request in English
 - Should we use other mathematical properties in describing AI
 - Distributive, Associative, communicative,



AI Research Notes

- AI treats all information equally - no weighting of proven information, one piece of information can pollute the repository
 - AI does not look at corroboration of evidence
 - Is AI looking at the first answer or sorting across answers - similar to traffic routing problems
- AI has no context, it may not have a memory of entire situation
 - A photo of salmon in a river shows salmon filets on the stream since most salmon on the web is related to cooking not swimming
 - A student had AI answer 14 questions to configure a network firewall - but nothing worked - the 14 commands were correct but independent and did not share a context
- Correlation does not mean causation - AI may assume causation
 - Serial logic issue



AI Serial logic

- An AI serial logic issue (also known as a sequential or chain-of-thought failure) happens when a model makes an error by following a logical sequence that sounds correct at each step, but fails to reach a correct or sensible conclusion. [[1](#), [2](#)]
- **The "Car Wash" Logic Trap**
- A widely studied example of this is the **Car Wash Test**. [[1](#)]
- **The Prompt:** "I want to wash my car. The car wash is 50 m away. Should I walk or drive?" []
- **The AI's Serial Process:**
 - The AI identifies the distance: 50 meters.
 - It evaluates travel methods for a short distance: "Walking is healthy and uses no fuel."
 - It concludes: "You should walk." [[1](#)]
- **The Logic Failure:** The AI sequentially processes the distance but entirely fails to account for the physical goal—**you cannot wash a car without the car being at the car wash**. By optimizing for "fuel economy" at each step, it ignores the foundational reality of the task. [[1](#)]



AI Research Notes

- How to decide among conflicting issues
 - AI hallucinations
 - Human Analogy - doctors are trained to look at multiple possibly conflicting issues to be able to make a determination
 - AI is goaled to make the reader happy – provide a solution
- Does AI provide alternatives or a single answer?
 - Could it provide courses of action with percentages
 - Let the human see alternatives to be able to fine tune questions
- Cyber security - how to verify that information has not been altered or poisoned
 - Develop secure information repositories
- Key determinations in when to use a tool
 - What is the role of AI
 - Researching Facts – provide understanding
 - What is the role of the human?
 - Making a determination?



Business Experience

- You can't get a job without experience, but how do you get experience?
- Get use to operating in the business world – Develop the mindset
 - Business Common Sense – Do students know what business does?
 - What is the basis of a business decision?
 - How does your boss and his/her boss think?
 - Practice
- Understand the entire business situation – not just the brief
 - Example: China – Solar Panels in Desert create Oasis – that is only a sound bite
 - Story is that they have to bring in sheep to maintain short grass that block panels
 - But why is grass getting tall in a desert? There is shade however:
 - They don't mention that they wash the panels every week adding water to the desert
 - They don't talk about scraping the land (destroy all plants/animals) to install them
 - What are the real cause and effects?
 - What are the unintended consequences?



Business Experience – Practice the Mindset

- Business Memos
 - Introduce Business format
 - Summary upfront – Big Look Up Front
- Verbal presentations – focus on discussions
- It is not the plan – it is the planning process
 - No scenario or situation will match the training situation
 - It is important to understand what you have and what will happen
 - Assets
 - Cause and Effects from decisions so you react to the changing situation

Communications

-



What is required in your memos

- Memo format
 - No need for salutation: Dear, To whom it may concern
- Business analysis
 - BLUF – Bottom Line Up Front
 - What was the analysis of the data –
 - Include graphics to support your analysis
 - Type of Memo
 - Information Memo - analysis
 - Action Memo -Courses of action (COA) – if there is a selection to be made
 - Usually 3 – one is do nothing different – consequences of doing the same thing
 - Then 2 alternatives
 - Background – techniques and issues



What is BLUF format?

- BLUF is a military communications acronym—it stands for “bottom line up front”—that's designed to enforce speed and clarity in reports and emails. The basic idea is simple: put the most important details first. Don't tease or delay your main point because people are busy and their time is valuable. It declares the purpose of the email and action required. The BLUF should quickly answer the five W's: who, what, where, when, and why, Sep 9, 2019
- BLUF: The Military Standard That Can Make Your Writing ...<https://www.animalz.co> › blog › bottom-line-up-front
- How do you write BLUF?
- In the BLUF format, the first sentence of each paragraph will sum up all of the information in the paragraph. A good BLUF should cover all of the information in a paragraph, like an umbrella.
 - An executive overview



How to Write Email with Military Precision

- by Kabir Sehgal - November 22, 2016
- **Summary.** When you send an email, the first thing your recipient sees is the subject line, so make sure it's as clear as possible: What is your email's purpose? What do you want your recipient to do? Take a page from military personnel. Their subject lines use keywords in...more
- In the military, a poorly formatted email may be the difference between mission accomplished and mission failure. During my active duty service, I learned how to structure emails to maximize a mission's chances for success. Since returning from duty, I have applied these lessons to emails that I write for my corporate job, and my missives have consequently become crisper and cleaner, eliciting quicker and higher-quality responses from colleagues and clients. Here are three of the main tips I learned on how to format your emails with military precision:
- **1. Subjects with keywords.** The first thing that your email recipient sees is your name and subject line, so it's critical that the subject clearly states the purpose of the email, and specifically, what you want them to do with your note. Military personnel use keywords that characterize the nature of the email in the subject. Some of these keywords include:
 - ACTION – Compulsory for the recipient to take some action
 - SIGN – Requires the signature of the recipient
 - INFO – For informational purposes only, and there is no response or action required
 - DECISION – Requires a decision by the recipient
 - REQUEST – Seeks permission or approval by the recipient
 - COORD – Coordination by or with the recipient is needed
- The next time you email your direct reports a status update, try using the subject line: INFO – Status Update. And if you need your manager to approve your vacation request, you could write REQUEST – Vacation. If you're a project manager who requires responses to your weekly implementation report from several people, type ACTION – Weekly Implementation Report. These demarcations might seem obvious or needlessly exclamatory because they are capitalized. But your emails will undoubtedly stand out in your recipient's inbox, and they won't have to work out the purpose of your emails. (It also forces you to think about what you really want from someone before you contribute to their inbox clutter.)

How to Write Email with Military Precision

- **2. Bottom Line Up Front (BLUF).** Military professionals lead their emails with a short, staccato statement known as the BLUF. (Yes, being the military, there is an acronym for everything.) It declares the purpose of the email and action required. The BLUF should quickly answer the five W's: who, what, where, when, and why. An effective BLUF distills the most important information for the reader. Here's an example BLUF from the Air Force Handbook:
- BLUF: Effective 29 Oct 13, all Air Force Doctrine Documents (AFDDs) have been rescinded and replaced by core doctrine volumes and doctrine annexes.
- The BLUF helps readers quickly digest the announcement, decision, and when the new procedures go into effect. The reader doesn't necessarily want to know all the background information that led to the decision. He or she likely wants to know "how does this email affect me?" and the BLUF should answer this question every time.
- For my corporate job, I don't use the acronym "BLUF" because it would be unclear to recipients, but I have started leading with "Bottom Line" in bold at the start of my notes. Sometimes, I even highlight the bottom line in yellow so that my point is abundantly clear. Here is an example of a BLUF adapted for corporate use:
 - Subject: INFO – Working from home
 - Shannon,
 - Bottom Line: We will reduce the number of days that employees can work from home from three to one day per week effective December 1st.
 - Background:
 - This is an effort to encourage team morale and foster team collaboration
 - All members of the management committee supported this decision
- Shannon knows that no response is required because it was marked INFO. She also quickly grasps the information in the email because of the Bottom Line. Because this is a big change in corporate policy, background details are provided to show that the decision is final, supported by management, and intended to result in positive effects for the company.



How to Write Email with Military Precision

- **3. Be economical.** Military personnel know that short emails are more effective than long ones, so they try to fit all content in one pane, so the recipient doesn't have to scroll. They also eschew the passive voice because it tends to make sentences longer, or as the Air Force manual puts it, "Besides lengthening and twisting sentences, passive verbs often muddy them." Instead, use active voice, which puts nouns ahead of verbs, so it's clear who is doing the action. By using active voice, you are making the "verbs do the work for you." Instead of, "The factory was bombed by an F18," military professionals would say, "An F18 bombed the factory."
- Even though short emails are usually more effective, long emails abound, even in the military. If an email requires more explanation, you should list background information after the BLUF as bullet points so that recipients can quickly grasp your message, like in the above example.
- Lastly, to prevent clogging inboxes, military professionals link to attachments rather than attaching files. This will force the recipient to check the website that has the attachment, which will likely provide the most recent version of a file. Also, the site will verify that the recipient has the right security credentials to see the file, and you don't inadvertently send a file to someone who isn't permitted to view it.
- Here is an email example for corporate use that uses keywords in the subject, bottom line, background bullets, and active voice:
 - Subject: INFO – Meeting Change
 - Shannon,
 - Bottom Line: We scheduled the weekly update meeting for Thursday at 2 PM CST to accommodate the CFO's schedule.
 - Background:
 - We searched for other available times, but this is the only time that works, and it's important that you are on the call, so that you can address your P&L.
 - CFO will be in Boston on Thursday meeting at an offsite with the management committee.
- He wants to review the financial report that can be found here (insert link) before the call.
- By adopting military email etiquette, you will introduce a kernel of clarity to your correspondence and that of your colleagues and clients.

Explain it to a 5 Year Old



- Have you ever been asked to simplify your complex work for someone entirely new to the concept?
- As a data engineer or data scientist, you may encounter a common interview question: “Can you explain data analysis as if you were speaking to a 5-year-old?”
- This question may seem lighthearted, but it holds significance.
- Why is it relevant?
 - 1. Testing Foundational Understanding: It evaluates how well you grasp the core principles of data analysis.
 - 2. Simplifying Complexity: It challenges you to explain intricate ideas in simple, relatable terms — a vital skill for any communicator or collaborator.

- Let's imagine explaining data analysis to a child using chocolates as an example.
- Step 1: Set the Scene
 - I would start by asking the child a fun question: "Which chocolate do you like the most: KitKat, Cadbury, or Hershey's?"
 - "The child might pause to think and then give an answer. It could be a single choice, or they might name multiple chocolates.
- Step 2: Follow Up with Questions
 - If they pick one chocolate: "Why do you like this chocolate more than the others?"
 - "If they pick multiple chocolates: "What is similar about these chocolates that makes you like them?"
 - "The child might respond with something like, "It has more nuts," or "It's crunchier," or even, "They taste the same."

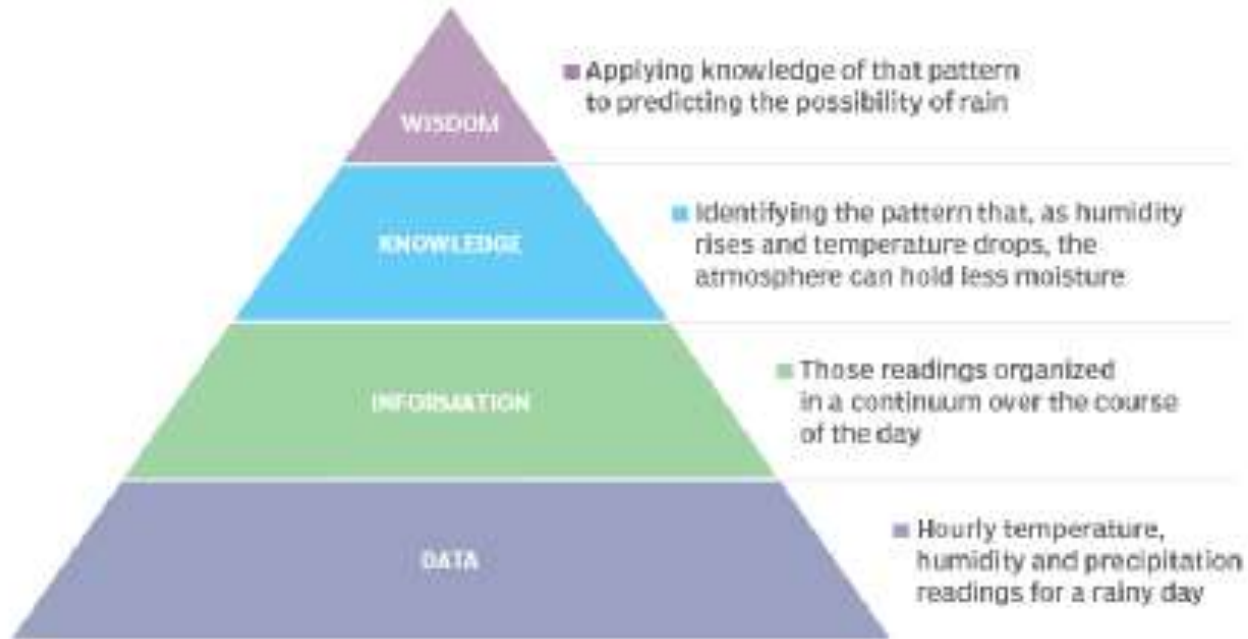
- Step 3: Simplify When Necessary

- If the child struggles to answer, I would simplify the questions further, like:
 - Which chocolate do you buy most often?
 - Which chocolate does your friend like or buy?
 - Have you tried other chocolate brands?
 - Did your parents suggest these chocolates to you?
 - What attracts you the most about these chocolates?
 - Do you like the wrapping or color of this chocolate more than others?
 - Through this series of questions, I'd gather clues to understand why the child prefers a particular brand (or brands) over others.

- Step 4: Connect It to a Bigger Picture
 - This process — asking questions, analyzing responses, and adjusting the approach to uncover insights — is the essence of data analysis. Businesses, like shopkeepers, use this technique to identify what their customers like (in this case, chocolates) and tailor their strategies to increase sales and profits.
- Key Takeaway
 - Data analysis isn't just about numbers or algorithms; it's about asking the right questions, understanding the patterns in answers, and making informed decisions. Whether it's a shopkeeper selling chocolates or a data scientist solving real-world problems, the principle is the same
- .And that's how I would explain data analysis to a 5-year-old — by making it relatable, simple, and fun!

Information

An example of data-information-knowledge-wisdom



What is information?



Information

- Information is stimuli that has meaning in some context for its receiver. When information is entered into and stored in a computer, it is generally referred to as data. After processing -- such as formatting and printing -- output data can again be perceived as information. When information is compiled or used to better understand something or to do something, it becomes knowledge.
- **Data** – discrete, raw facts about a given situation with no analysis or interpretation applied
- **Information** – applying description and meaning to data to make it useful
- **Knowledge** – Information that has insight, content and a frame of reference applied so it can be interpreted.
- **Wisdom**- Knowledge is converted into wisdom by applying judgment and action to the information



Data

- Data is collected from a variety of sources, such as computers, sensors and devices. It is typically used in business, science and engineering. Data is often presented in the form of numbers, but it can also come as text, visuals, graphics and sounds. Data can also be analyzed and used to create information that could not be obtained by just looking at the original data.

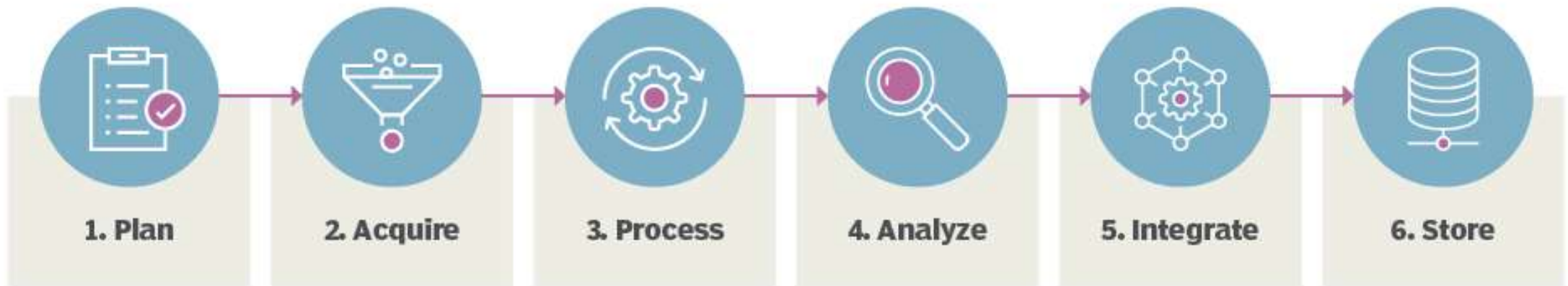


- The most common types of data in data science are the following:
- **Quantitative data** is numerical data, or data that can be expressed mathematically. Discrete and continuous data are types of quantitative data.
- **Qualitative data** is data that cannot be measured, counted or easily expressed with numbers. It is data that comes from text, audio or images. It can be shared using data visualization tools, such as timelines, infographics and word clouds.
- **Nominal data** is the simplest form of data in statistics. It is data that is used to name or label a variable; it isn't used to measure things or put them in any order. Examples of nominal data include ethnicity, gender, eye color.
- **Ordinal data** is data that takes on values within a known range and follows a natural order. A common example of ordinal data is income levels where incomes are ranked in specific ranges, such as \$0-\$50K, \$50K-\$75K, \$75K-\$100K, etc. The purpose of ordinal data is to rank items in order of priority or value. The numbers are not used for calculations.
- **Discrete data**, also called categorical data, is data that is divided into discrete categories, or groups, that are distinctly different from each other. With discrete data, only a specific number of values are possible, and those values cannot be subdivided. For example, the number of people a company employs is a discrete data point.
- **Continuous data** is a term used to describe data that is measurable and observable in real time. It can be measured on a scale or a continuum and subdivided into finer values. Continuous data is often recorded at set intervals and then analyzed using statistical software. The amount of time it takes to complete a task is an example of continuous data.

What is the data processing cycle?

- The data processing cycle is the framework that data center managers use to make data accessible and useful to users. It is a portion of the data lifecycle. Data enters the data center where it is processed, and then it is sent to the user who makes use of it in a business application.
- The part of the data lifecycle referred to as the data processing cycle is divided into the following three stages:
 1. This is the stage where data is collected from multiple sources -- point-of-sale locations, call centers and sensors, for example.
 2. The data is sorted, organized, cleansed and entered into a database or system. It is then transformed into a format that users can understand and make use of.
 3. The newly processed and transformed data is sent to users or stored in a way that they will have access to it when needed.

Traditional data lifecycle





Converting data to information

- Data and information are not the same. Data refers to numerical and qualitative observations. Information is created when data is presented in a way that has meaning to the recipient. To turn data into information, it must be processed and organized. Presenting data in a way that has meaning and value is called information design, and it is an important field in both Information architecture and human-computer interaction.
- Five characteristics of data quality and high-quality information in a database include the following:
 1. Information must come from a reliable source of information.
 2. Information cannot be partial or have details missing.
 3. Mechanisms must be in place to ensure that new data doesn't contradict existing data.
 4. Information must be distinctive and add value to a database.
 5. Information in a database must be timely and up to date.



Converting information to knowledge and wisdom

- Knowledge is information that has been processed, analyzed and interpreted, and can be used to make decisions. The concept of knowledge involves not just the information, but the ability to access it, as well. For example, most applications, including models and simulations, include a form of stored knowledge.
- Wisdom is the synthesis of information, knowledge and experience in a way that applies knowledge to real-life situations. The concept of wisdom enables the understanding of patterns and their driving factors. It ultimately enables the prediction of future events.
- Artificial intelligence (AI) has enabled computers to learn, problem-solve and perform tasks that usually require human intelligence. These technologies enable computers to take actions based on what the data provided indicates is the best course of action. AI is used in expert systems to diagnose disease, buy and sell stock and play chess better than a human. However, IT has not yet attained a level of human wisdom.



Data Life Cycle

- The data life cycle is the sequence of stages that a particular unit of data goes through from its initial generation or capture to its eventual archival and/or deletion at the end of its useful life.
- Although specifics vary, data management experts often identify six or more stages in the data life cycle. Here's one example:
- **Generation or capture:** In this phase, data comes into an organization, usually through data entry, acquisition from an external source or signal reception, such as transmitted sensor data.
- **Maintenance:** In this phase, data is processed prior to its use. The data may be subjected to processes such as integration, scrubbing and extract-transform-load (ETL).
- **Active use:** In this phase, data is used to support the organization's objectives and operations.
- **Publication:** In this phase, data isn't necessarily made available to the broader public but is just sent outside the organization. Publication may or may not be part of the life cycle for a particular unit of data.
- **Archiving:** In this phase, data is removed from all active production environments. It is no longer processed, used or published but is stored in case it is needed again in the future.
- **Purging:** In this phase, every copy of data is deleted. Typically, this is performed on data that is already archived.
- Data lifecycle management (DLM) is becoming increasingly important since the explosion of big data and the ongoing development of the Internet of Things (IoT). Enormous volumes of data are being generated by an ever-increasing number of devices all over the world. Proper oversight of data throughout its life cycle is essential to optimize its usefulness and minimize the potential for errors. Finally, archiving or deleting data at the end of its useful life ensures that it does not consume more resources than necessary.



What is Data Science

- Data science is a multidisciplinary field focused on finding actionable insights from large sets of raw and structured data. The field primarily focuses on providing answers to the things that are not known.
- This field concentrates on using different techniques to obtain answers, incorporating computer science, predictive analytics, statistics, and machine learning to dissect through massive datasets in an effort to establish solutions to problems that haven't been thought of yet. The main goal in this field is to ask questions and locate potential avenues of study, with less concern for specific answers and more emphasis placed on finding the right question to ask. Experts accomplish this by predicting potential trends, exploring disparate and disconnected data sources, and finding better ways to analyze information.
- What is Data Analytics?
- Data analytics focuses on processing and performing statistical analysis of existing datasets. This field concentrates on creating methods to capture, process, and organize data to uncover actionable insights for current problems, and establishing the best way to present this data. Data analytics also encompasses a few different branches of broader statistics and analysis which help combine different sources of data and locate connections while simplifying the results.



What is Data Science

- What Is the Difference?
- While many people use the terms interchangeably, data science and big data analytics are unique fields, with the major difference being the scope. Data science is an umbrella term for a group of fields that are used to mine large datasets. Data analytics is a more focused version of this and can even be considered part of the larger process.
- Another significant difference between the two fields is a question of exploration.
- Data science isn't concerned with answering specific queries, instead parsing through massive datasets in sometimes unstructured ways to expose insights. Data analysis works better when it is focused, having questions in mind that need answers based on existing data.
- Data science produces broader insights that concentrate on which questions should be asked, while big data analytics emphasizes discovering answers to questions being asked.
- More importantly, data science is more concerned about asking questions than finding specific answers. The field is focused on establishing potential trends based on existing data, as well as realizing better ways to analyze and model data.



Data Science



VS

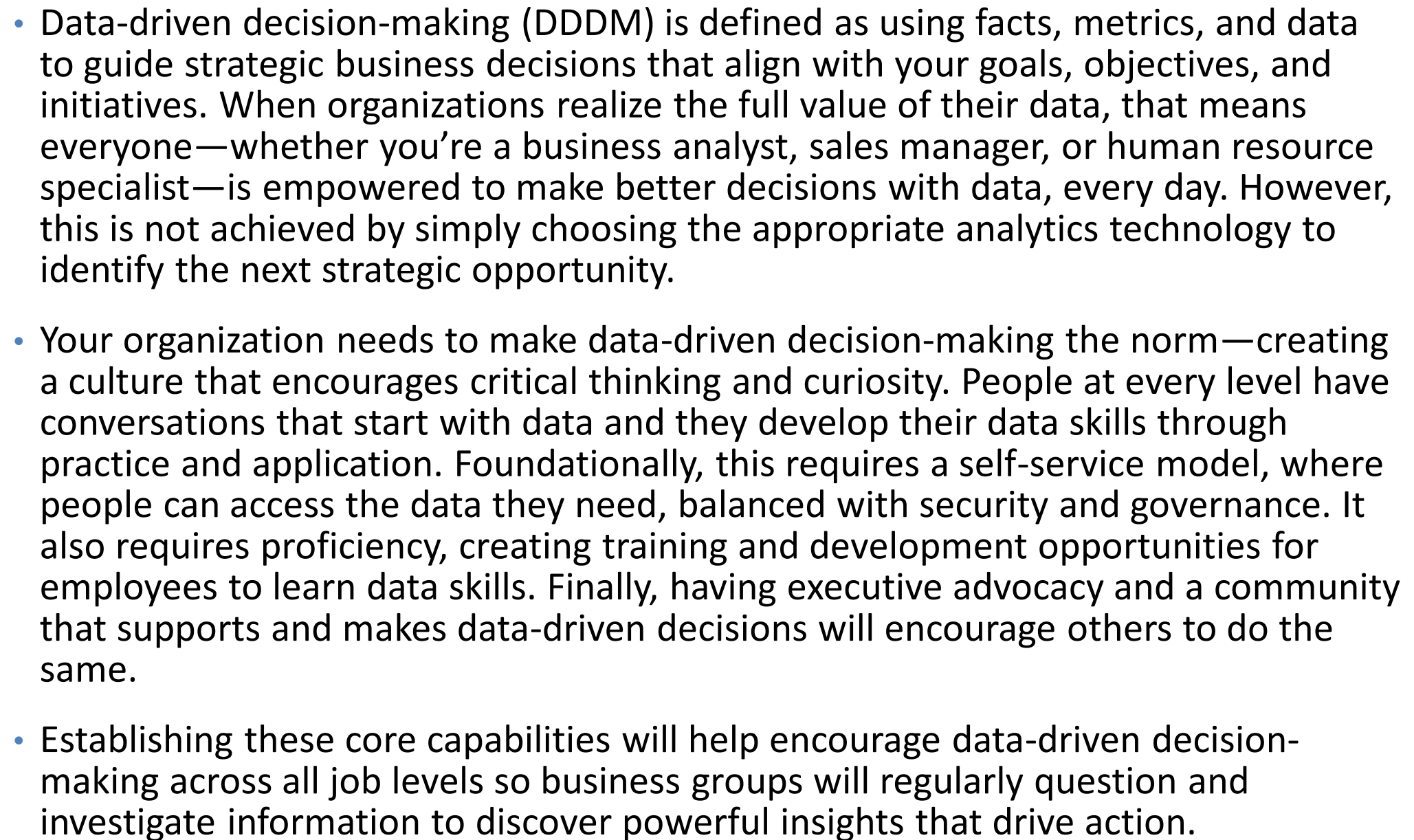


Data Analytics



So what is a decision?

- the act or process of deciding. the moment of decision has come.
- : a determination arrived at after consideration :
- So what is data driven decision making?





- **The importance of data-driven decision-making**
- The amount of information collected has never been greater, but it's also more complex. This makes it difficult for organizations to manage and analyze their data. In fact, NewVantage Partners recently reported that 98.6 percent of executives indicate that their organization aspires to a data-driven culture, while only 32.4 percent report having success. A 2018 IDC study also noted that organizations have invested trillions of dollars to modernize their business, but 70 percent of these initiatives fail because they prioritized technology investments without building a data culture to support it.
- In pursuit to be data-driven, many enterprises are developing three core capabilities: data proficiency, analytics agility, and community. Transforming how your company makes decisions is no easy task, but incorporating data and analytics into decision-making cycles is how you will see the most transformative impact on your organization. This level of transformation requires a dedicated approach to developing and refining your analytics program.



- **Organizations benefiting from data-driven decision-making**
- Thanks to modern business intelligence, organizations are inching closer and closer to understanding the value of data-driven decision-making across all departments and roles. Here are a few examples of organizations that are effectively leveraging the value of their people and their data.
- Lufthansa group increased organizational efficiency by 30%
- Providence St. Joseph Health improved quality measures and cost-of-care
- Charles Schwab Corporation increased their speed to business insights



Modern Data Analysis

- **AI-powered analysis.**
- Artificial intelligence (AI) and machine learning (ML) are resetting the expectations of modern analytics. Augmented analytics suggests new insights and connections and this helps you quickly analyze your data, increase your productivity, and make better, data-driven decisions.
- **ML, predictive, and prescriptive analytics.**
- Machine learning is the process of creating models from historical data in order to make future predictions. Automated machine learning allows you to leverage the power of predictive and prescriptive analytics in more user cases not handled by data scientists.
- **Data visualization, reports and dashboards**
- Stakeholders have an easier time understanding and collaborating on data when it's visualized in the form of charts, graphs and maps. Interactive digital dashboards and reports let you share your findings and allow all users to easily interact with the data.
- **Embedded analytics**
- Embedding data analysis capabilities within other applications, products, portals, workflows or processes helps people find insights and make better decisions faster. This is because rather than switching to a separate analytics application, all stakeholders can quickly access data and insights right within their workflows.



Data Driven Decision Making

- **Summary**

- Data-driven decision making is the process of collecting data based on your company's key performance indicators (KPIs) and transforming that data into actionable insights. This process is a crucial element of modern business strategy. In this article, we'll discuss the benefits of data-driven decision making and provide tips so you can make informed decisions at work.
- If there's a looming decision ahead of you at work, it's often hard to know which direction to go. If you go with your gut feeling, you may feel more confident in your choices, but will those choices be right for your team members? When you use facts to make decisions, you can feel more at ease knowing your choices are based on data and meant to maximize business impact.
- Whether outshining competitors or increasing profitability, data-driven decision making is a crucial part of business strategy in the modern world. Below, we dive into the benefits of data-driven decision making and provide tips for making these decisions at work.



What is data-driven decision making (DDDM)?

- Data-driven decision making is the process of collecting data based on your company's key performance indicators (KPIs) and transforming that data into actionable insights.
- Data-drive decision making (DDDM) is the process of collecting data based on your company's key performance indicators (KPI's) and transforming that data into actionable insights
- You can use business intelligence (BI) reporting tools during this process, which make big data collection fast and fruitful. These tools simplify data visualization, making data analytics accessible to those without advanced technical know-how.



What does being data-driven mean?

- In short, the concept of being data-driven refers to using facts, or data, to find patterns, inferences, and insights to inform your decision making process.
- Essentially, being data-driven means that you try to make decisions without bias or emotion. As a result, you can ensure that your company's goals and roadmap are based on evidence and the patterns you've extracted from it, rather than what you like or dislike.

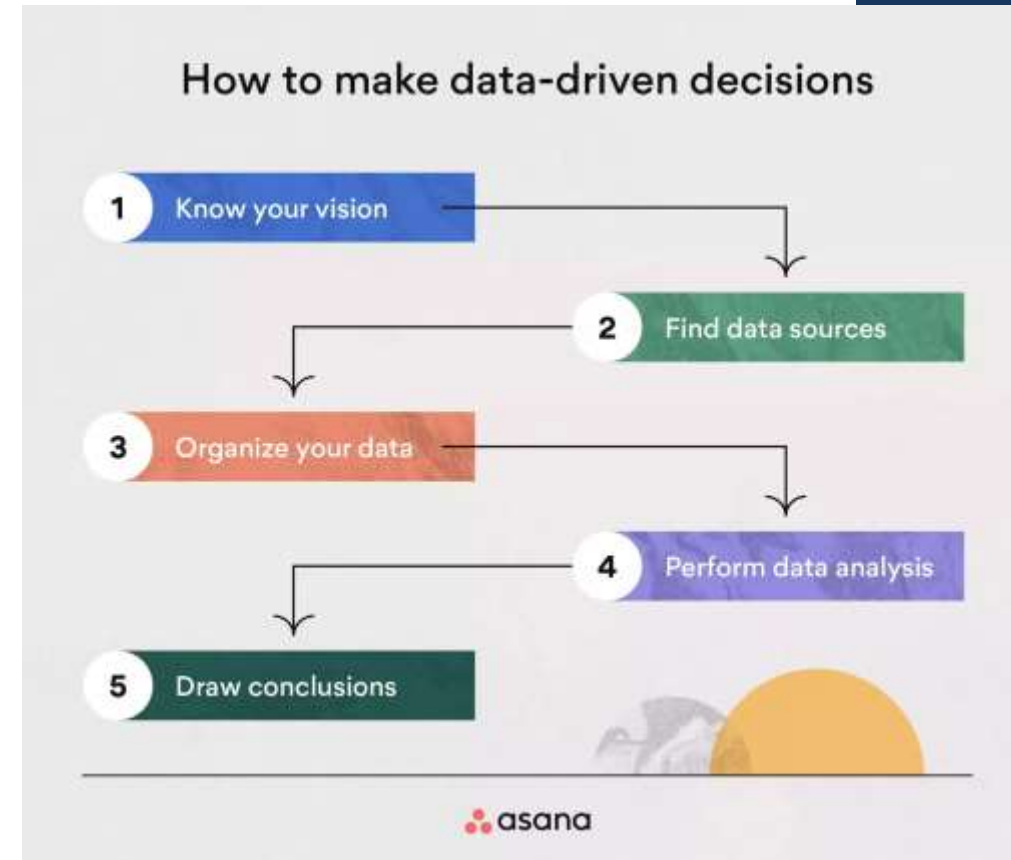


Why is data-driven decision making important?

- Data-driven decision making is important because it helps you make decisions based on facts instead of biases. If you're in a leadership position, making objective decisions is the best way to remain fair and balanced.
- The most informed decisions stem from data that measure your [business goals](#) and populates in real time. You can aggregate the data you need to see patterns and make predictions with [reporting software](#).
- Some decisions you can make with support from data include:
 - How to drive profits and sales
 - How to establish good management behavior
 - How to optimize operations
 - How to improve team performance
- While not every decision will have data to back it up, many of the

5 steps for making data-driven decisions

- Making data-driven decisions takes practice. If you want to improve your leadership skills, then you'll need to know how to turn raw data into actionable steps that work toward your company initiatives. The following steps can help you make better decisions when analyzing data.



1. Know your vision

- Before you can make informed decisions, you need to understand your company's vision for the future. This helps you use both data and strategy to form your decisions. Graphs and figures have little meaning without context to support them.
- Tip: Use your company's yearly objectives and key results (OKRs) or quarterly team KPIs to make data-backed decisions.



2. Find data sources

- Once you've identified the goal you're working towards, you can start collecting data.
- The tools and data sources you use will depend on the type of data you're collecting. If your goal is to analyze data sets pertaining to internal company processes, use a universal reporting tool. Reporting tools offer a single point of reference for keeping track of how work across your organization is progressing. Some reporting tools like Microsoft's Power BI let you gather data from various external sources. If you want to analyze marketing trends or competitor metrics, you can use one of those tools.
- Some general success metrics you may want to measure include:
 - Gross profit margin: Gross profit margin is measured by subtracting the cost of goods sold from the company's net sales.
 - Return on investment (ROI): The ratio between the income and investment, ROI is commonly used to decide whether or not an initiative is worth investing time or money in. When used as a business metric, it often tracks how well an investment is performing.
 - Productivity: This is the measurement of how efficiently your company is producing goods or services. You can calculate this by dividing the total output by the total input.
 - Total number of customers: This is a simple but effective metric to track. The more paid customers, the more money earned for the business.
 - Recurring revenue: Commonly used by SaaS companies, this is the amount of revenue generated by all of your current active subscribers during a specific period. It's commonly measured either monthly or annually.
- You can measure a variety of other data sets based on your job role and the vision you're working toward. Machine learning makes aggregating real time data simpler than ever before.
- Tip: Try to create a connected story through these metrics. If revenue is down, look at productivity and see if you can draw a connection. Keep digging through these metrics until you find a "why" for whatever problem you're trying to solve.

3. Organize your data

- Organizing your data to improve data visualization is crucial for making effective business decisions. If you can't see all your relevant data in one place and understand how it connects, then it's difficult to ensure you're making the most informed decisions.
- Tip: One way to organize your data is with an executive dashboard. An executive dashboard is a customizable interface that usually comes as a feature of your universal reporting tool. This dashboard will display the data that's most critical to achieving your goals, whether those goals are strategic, tactical, analytical, or operational.

4. Perform data analysis

- Once you've organized your data, you can begin your data-driven analysis. This is when you'll extract actionable insights from your data that will help you in the decision-making process.
- Depending on your goals, you may want to analyze the data from your executive dashboard in tandem with user research such as case studies, surveys, or testimonials so your conclusions include the customer experience.
- Does your team want to improve their SEO tools to make it more competitive with other options on the market? The data sets you can use to determine necessary improvements may include:
 - Competitors' performance data
 - Current SEO software performance data
 - Current customer satisfaction data
 - User research on a variety of SEO/marketing tools
- While some of this information will come from your organization, you may need to obtain some of it from external sources. Analyzing these data sets as a whole can be helpful because you'll draw a different conclusion than you would if you were to analyze each data set individually.
- Tip: Share your analytics tools with your whole team or organization. Just like any collaborative effort, data analysis is most effective when viewed from many perspectives. While you may notice one pattern in the data, it's entirely possible a teammate may see something completely different.

5. Draw conclusions

- As you perform your data analysis, you'll likely begin to draw conclusions about what you see. However, your conclusions deserve their own section because it's important to flesh out what you see in the data so you can share your findings with others.
- The main questions to ask yourself when drawing conclusions include:
 - What am I seeing that I already knew about this data?
 - What new information did I learn from this data?
 - How can I use the information I've gained to meet my business goals?
- Once you can answer these questions, you've successfully performed data analysis and should be ready to make data-driven decisions for your business.
- Tip: A natural next step after data analysis is writing down some SMART goals. Now that you've dug into the facts, you can establish achievable goals based on what you've learned.



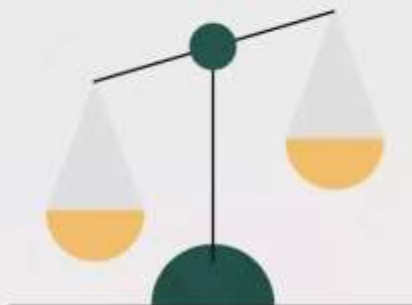
Data-driven decision making example

- While the data analysis itself happens behind the scenes, the way data-driven decisions affect the consumer is very apparent. Some examples of data-driven decision making in different industries include:
- E-commerce
 - Have you ever been shopping online and wondered why you're getting certain recommendations? Well, it's probably because you bought something similar in the past or clicked on a certain product.
 - Online marketplaces like Amazon track customer journeys and use metrics like click-through rate and bounce rate to identify what items you're engaging with most. Using this data, retailers are able to show you what you might want without you having to search for it.
- Finance
 - Financial institutions use data in a multitude of different ways, ranging from assessing risk to customer segmentation. Risk is especially prevalent in the financial sector, so it's important that companies are able to determine the risk factor before making any significant decisions. Historical data is the best way to understand potential risks, threats, and the likelihood they occur.
 - Financial institutions also use customer data to determine their target market. By grouping consumers based on socioeconomic status, spending habits, and more, financial companies can infer what consumers have the greatest lifetime value and target them.
- Transportation
 - Data science additionally plays a huge role in determining safe transportation. The U.S. Department of Transportation's Safety Data Initiative underscores the role that data plays in improving transportation safety.
 - The report pulls data from all types of motor crashes and evaluates factors like weather and road conditions to discover the source of problems. Using the hard facts, the department can work toward implementing more safety measures.

Benefits of data-driven decision making



Make confident
decisions



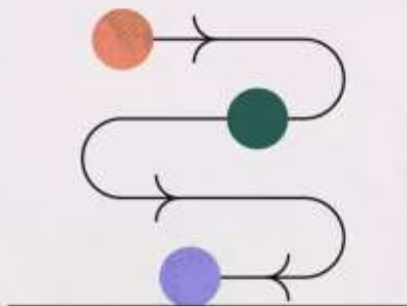
Guard against
biases



Find unresolved
questions



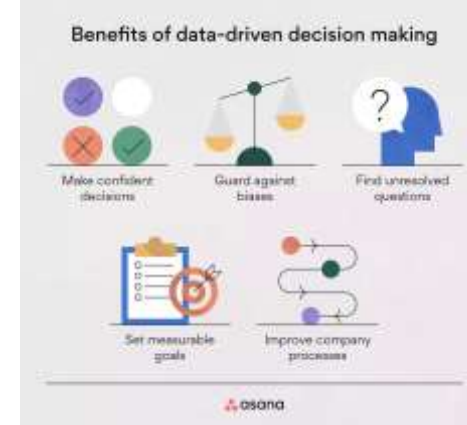
Set measurable
goals



Improve company
processes

Make confident decisions

- The businesses that outlast their competitors do so because they're confident in their ability to succeed. If the decision-makers within a business waiver in their choices, it can lead to mistakes, high team member turnover, and poor risk management.
- When you use data to make the most important business decisions, you'll feel confident in those decisions, which will push you and your team forward. Confidence can lead to higher team morale and better performance.
- **Guard against biases**
- Using data to make decisions will guard against any biases among business leaders. While you may not be aware of your biases, having internal favoritism or values can affect the way you make decisions.
- Making decisions directly based on the facts and numbers keeps your decisions objective and fair. It also means you have something to back up your decisions when team members or stakeholders ask why you chose to do what you did.



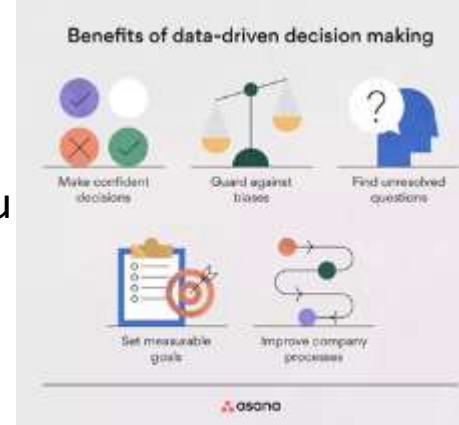


- **Find unresolved questions**

- Without using data, there are many questions that go unanswered. There may also be questions you didn't know you had until your data sets revealed them. Any amount of data can benefit your team by providing better visualization into areas you can't see without statistics, graphs, and charts.
- When you bring those questions to the surface, you can feel confident knowing your decisions were made by considering every bit of relevant information.

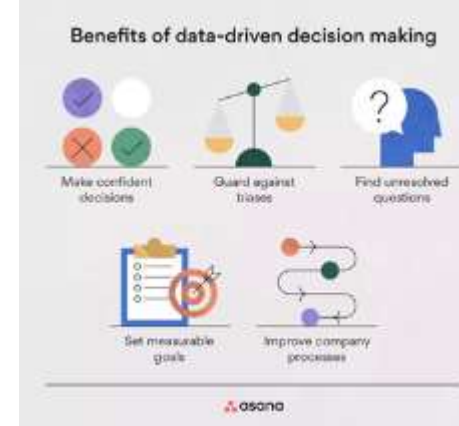
- **Set measurable goals**

- Using data is one of the simplest ways to set measurable goals for your team and successfully meet those goals. By looking at internal data on past performance, you can determine what you need to improve and get as granular as possible with your targets. For example, your team may use data to identify the following goals:
 - Increase number of customers by 20% year over year
 - Reduce overall budget spend by \$20,000 each quarter
 - Reduce project budget spend by \$500
 - Increase hiring by 10 team members each quarter
 - Reduce cost per hire by \$500
- Without data, it would be difficult for your company to see where they're spending their money and where they'd like to cut costs. Setting measurable goals ultimately leads to data-driven decisions because once these goals are set, you'll determine how to reduce the overall budget or increase the number of customers.





- **Improve company processes**
- There are ways to improve company processes without using data, but when you observe trends in team member performance using numbers or analyzing company spending patterns with graphs, the process improvements you make will be based on more than observation alone.
- Processes you can improve with data may include:
 - Risk management based on financial data
 - Cost estimation based on market pricing data
 - Team member onboarding based on new hire performance data
 - Customer service based on customer feedback data
- Changing a company process can be difficult if you aren't sure about the result, but you can be confident in your decisions when the facts are in front of you.

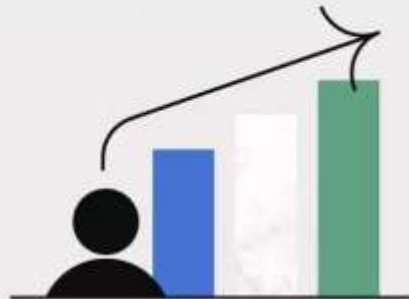


Tips for becoming more data-driven



Find the story

When analyzing data, look for patterns and meanings behind the numbers and figures.



Consult the data

Before making any gut decisions, see if the facts align with your feelings.



Learn data visualization

Make sure your data is visually appealing and easy to understand.



- **Find the story**

- The key to analyzing data, numbers, and charts is to look for the story. Without the “why,” the data itself isn’t much help, and the decision process is far more difficult. If you’re trying to become more data-driven in your decision making, look for the story the data is telling. This will be integral in making the right decisions.

- **Consult the data**

- Before making any organizational decision, ask yourself: Does the data support this? Data is everywhere and can be applied to any major decision. So why not consult it when making tough choices? Data is so helpful because it’s naturally void of bias, so make sure you’re consulting the facts before any decision.

- **Learn data visualization**

- Finding the story behind the data becomes easier when you’re able to visualize it clearly. While learning how to visualize data is often the toughest aspect of establishing a data-driven culture, it’s the best way to recognize patterns and discrepancies in the data.
- Familiarize yourself with different tools and techniques for data visualization. Try to get creative with the different ways to present data. If you’re well-versed in data visualization, your data storytelling skills will skyrocket.





Understanding the Issue at hand

- What are the key points of data?
 - When is a piece of information a sound byte – does it tell all of the story?
- Do you know them or are you exploring – supporting a decision vs exploration
- California drought
 - Is there a drought?
 - Where is the water stored?
 - How do you measure the snow pack
 - When do you measure the snow pack
- Climate Change
 - Is it a normal pattern of change?
 - We have had mini ice ages in the past
 - What is the source of the data?
 - Do we have enough data?
- Compare Cars
 - Gas vs EV
 - Commuter vs gas guzzler



Be willing to listen to both sides

- Each issue will have multiple viewpoints?
- Your viewpoint is your version of reality
- Address the key issues of the alternate opinions
- Do you need emotion?
- IBM Sales Model – part of sales / support training
 - Introduction – develop a relationship
 - People only buy from people that they like
 - Court cases are judged on the jury's opinions of the attorneys
 - Propose solutions
 - **Identify and address objections**
 - Ask for the order – if I meet your objections will you order
 - Close it



TED Talks – Big Data

- **Tim Smith**
- https://www.ted.com/talks/tim_smith_big_data
- There is a mind-boggling amount of data floating around our society. Physicists at CERN have been pondering how to store and share their ever more massive data for decades- stimulating globalization of the internet along the way, whilst 'solving' their big data problem. Tim Smith plots CERN's involvement with big data from fifty years ago to today. [Directed by Biljana Labovic, narrated by Tim Smith, music by Corey Allen Jackson].
- 6 minutes



TED Talks – Big Data is Better Data

- Kenneth Cukier
- Self-driving cars were just the start. What's the future of big data-driven technology and design? In a thrilling science talk, Kenneth Cukier looks at what's next for machine learning -- and human knowledge.
- https://www.ted.com/talks/kenneth_cukier_big_data_is_better_data?language=en
- 15 minutes



Discussion

- Is Coca Cola Bad for you?
 - It is acid
 - It can be used to clean battery terminals in your car
 - It has too much sodium
- Even one or two colas a day could increase your risk of type 2 diabetes by more than 20%. Sugar intake is linked to high blood pressure, high cholesterol, and excess fat, all of which increase the risk of heart disease. Colas and other sugary drinks have been linked to an increased risk of pancreatic cancer – Web MD
- So why do people consume it?
- How would you analyze it?
 - What data would you want?
- How would you compare it to smoking or drinking alcohol?



Business Prediction Errors....

- It's scary how often ideas or concepts are rejected. History is littered with examples of people who wrote off a piece of technology, or said someone wouldn't amount to anything, and later had to pull their foot from their mouth.
- In fact, some predictions are so far from the mark, they're incredible. Here are eight embarrassingly bad business predictions that made the originator eat crow later on. A lot of crow.
- **1. Steve Ballmer on the iPhone's failure**
- "[Apple's iPhone] is the most expensive phone in the world and it doesn't appeal to business customers because it doesn't have a keyboard which makes it not a very good e-mail machine..."
- Steve Ballmer, Microsoft's CEO, uttered this famous quote in 2007. Needless to say, he missed the mark entirely; today, Apple is the top smartphone vendor in the world. Business customers are buying it and using it for e-mail among many, many other things.
- Apple's famous touch screen also became one of the biggest selling points of the phone, proving Mr. Balmer wrong yet again.



Prediction Errors....

- **2. Business Week on the failure of Japanese cars in the U.S.**
 - "With over 50 foreign cars already on sale here, the Japanese auto industry isn't likely to carve out a big slice of the U.S. market."
 - Those famous words were published in the prominent business magazine on August 2, 1968.
 - In case you haven't noticed, Japanese automakers have done very well in the U.S. In fact, 2008 was a particularly strong year for Japanese auto manufacturers, as Toyota became the world's biggest car manufacturer.
- **3. Decca Records on the future of The Beatles**
 - "We don't like their sound, and guitar music is on the way out," said Decca Records about The Beatles in 1962.
 - Little did the record company know, The Beatles would go on to be the best-selling band in history. Oh, and guitar music was definitely not on the way out. Two strikes for Decca.



Prediction Errors....

- **4. One publisher's take on J.K. Rowling's Harry Potter**
- "Children just aren't interested in witches and wizards anymore," an anonymous publishing executive told J.K. Rowling in 1996.
- Rowling, of course, went on to write one of the most-beloved children's series of all-time, selling more than 400 milion copies of stories about witches, wizards and goblins.
- **5. IBM on the need for photocopiers**
- "The world potential market for copying machines is 5,000, at most," said IBM to Xerox in 1959.
- After releasing the first plain paper photocopier in 1959, Xerox had revenues in \$60 million in 1961, and by 1965 it reached revenues of over \$500 million. In 2010, the photocopy giant made \$21.6 billion in revenue.



Prediction Errors....

- **6. Movie producer Darryl Zanuck on the future of television**
- "Television won't last because people will soon get tired of staring at a plywood box every night," said Zanuck to 20th Century Fox in 1946.
- Since Mr. Zanuck's shortsighted prediction, television (and Fox) have done quite well for themselves. In 2009, Nielsen found that over half of U.S. homes have three or more TVs.
- It turns out people do want to stare at the tube every night.
- **7. American engineer Ken Olson on the need for the personal computer**
- "There is no reason anyone would want a computer in their home," said Olson, co-founder of Digital Equipment Corp., in 1977.
- While the concept of the home computer has changed since 1977, those words couldn't be farther from the mark. In 2005, hard drive manufacturer Seagate performed a study that found that 76 percent of Americans had a personal computer. Six years later, of course, that number has most likely risen, along with the number of personal computers per household.



Prediction Errors....

- **8. A Yale University professor on the demand for overnight parcel delivery**
- "The concept is interesting and well-formed, but in order to earn better than a 'C', the idea must be feasible."
- This was the response of a Yale University professor to Fred Smith, on his paper proposing reliable overnight delivery service. Smith would later found FedEx.
- Smith not only created demand for overnight package delivery, but he built a \$34.7-billion empire.
- If there's one thing history has taught us, it is that we can't predict the future. We don't know what technologies will take hold, what discoveries will be made or what pop culture will evolve into. So don't listen to the experts.

Does Ice Cream or Swimming Cause Polio?

- "Correlation is not causation" means that just because two things correlate does not necessarily mean that one causes the other. As a **seasonal** example, just because people in the UK tend to spend more in the shops when it's cold and less when it's hot doesn't mean cold weather causes frenzied high-street spending.
- A correlation between variables, however, does not automatically mean that the change in one variable is the cause of the change in the values of the other variable. Causation indicates that one event is the result of the occurrence of the other event; i.e. there is a causal relationship between the two events.
- For example, the more fire engines are called to a fire, the more damage the fire is likely to do. In this case, the damage is not a result of more fire engines being called. In fact, both variables (the number of fire engines and the amount of damage done) are caused by the size of the fire.
- Even if there *is* a causal relationship between variables, it can be difficult to tell the direction of the relationship – which variable causes the other to change? For example, there might be a correlation between people's mood and their physical health, but it is not obvious which variable influences the other – do good moods improve physical health, or does good physical health improve people's moods?
- They noticed that the number of polio cases rose in the summer. They also noticed that people eat more ice cream in the summer. So... they declared that **ice cream was the problem!** Parents were told not to feed their children ice cream, and sales plummeted!
- Fearful parents jumped on this association, and began warning their children against swimming in pools, lakes, or any area with open water. Pools were vacated, lake homes were avoided, and swimming became seen as a dangerous exercise.



Data Mining – Are you sure?

- Experts have called 2017 the year of data literacy and digital transformation. While data is a key component that drives true digital transformation, too often companies approach data and analytics projects the wrong way. In fact, a mere 13% of data and analytics projects reach completion, and of those that do, only 8% of company leadership report being completely satisfied with the outcome.
- Why are data science project results so dismal?
- Most failures can be traced back to four major pitfalls: starting with the wrong questions; using faulty data; weak stakeholder buy-in; and lack of diverse expertise.
- Recognizing these common hazards upfront puts CIOs and IT Directors in a much better position to lead data science projects that drive valuable insights, and contribute to an organization's overall successful digital transformation.



Data Mining – Common Hazards

- **1. Starting with the wrong questions.**

- All too often data science projects begin by analyzing data with the expectation that an interesting insight will reveal itself and become the basis for the business case that justifies the transformation.
- This "exploratory analysis" approach often generates dozens of potential data projects that could yield compelling results. But do any of them produce a strong business case that can, for example, reduce costs, encourage repeat customers or retain employees? The scope of an "exploratory" project is simply too broad to drive useful analysis and is a waste of IT resources.
- The better approach is to initiate the project with an established goal that maps directly to creating business value. Projects that follow the "hypothesis-testing" approach begin with a specific set of clearly-defined questions that indicate which data should be analyzed.
- This targeted approach streamlines the data mining and analysis process by pairing business justification with business action, thereby directing IT resources to the information most likely to produce credible and meaningful findings. Starting with the right question sets the stage for a successful data science project through increased accuracy and efficiency, resulting in purposeful insight.

Data Mining – Common Hazards

- **2. Using faulty data.**

- Using accurate data is fundamental to a project's success, but bad data is the most underestimated cause of failure. Oftentimes companies simply do not spend enough time cleansing data. Because of the criticality of clean data, a good guideline is to allocate 80% of the projected project timeline to data clean-up.
- While this may seem excessive, doing a thorough job becomes that project's most significant time saver since working with clean data expedites all subsequent steps. Consider that even a simple error will produce a faulty insight with the potential to sink an entire project and cause leadership to withdraw support for future digital transformation initiatives.
- Modern cloud and data ingestion tools facilitate the consolidation of unstructured data, which can then be pulled out, mined and correlated in different ways, making data easier to manage while reducing time, infrastructure and errors down the line.

Data Mining – Common Hazards

- **3. Weak stakeholder buy-in.**

- Data science projects may impact business leaders across the company. Without stakeholder support and commitment to implement changes, projects could be stalled or fail.
- The best way to ensure business alignment across the organization is to produce a solid data strategy and roadmap to keep everyone on track.
- Stakeholders must believe in the project's purpose and commit to following through on implementation in their department when the time comes. When stakeholders see the value in an initiative supported by a solid business case, the likelihood of project failure caused by a stakeholder barrier greatly diminishes.

Data Mining – Common Hazards

- **4. Lack of diverse expertise.**
- It's a common misconception that any project involving data should be the sole responsibility of the IT department. This short-sighted view couldn't be more dangerous to a project's success. Using the right data analysis tools is important, but the output falls short of offering meaningful and useful insight. Involving the right human talent in the project is essential, irrespective of the department they ascribe to.
- The most successful data science projects employ team members across departments with diverse skillsets, including an understanding of quantitative research, statistics and subject matter expertise dependent on the initiative's question and focus.
- Together, they can bring different perspectives, proficiencies and experience to mold the project's objectives and direction as it progresses.
- Data projects also benefit from having someone on the team who understands the internal workings of the business to ensure it remains aligned with original business goals. More eyes on the project up the chances that mistakes will be caught, while leveraging the collective knowledge and talents of the entire team.

Major issues in Data Mining :

- **Mining different kinds of knowledge in databases –**
- The need for different users is not same. Different users may be interested in different kinds of knowledge. Therefore it is necessary for data mining to cover a broad range of knowledge discovery tasks.
- **Interactive mining of knowledge at multiple levels of abstraction –**
- The data mining process needs to be interactive because it allows users to focus on search for patterns, providing and refining data mining requests based on returned results.
- **Incorporation of background knowledge –**
- To guide discovery process and to express discovered patterns, background knowledge can be used to express discovered patterns not only in concise terms but at multiple levels of abstraction.
- **Data mining query languages and ad-hoc data mining –**
- Data Mining Query language that allows user to describe ad-hoc mining tasks should be integrated with a data warehouse query language and optimized for efficient and flexible data mining.



- **Presentation and visualization of data mining results –**
- Once patterns are discovered it needs to be expressed in high-level languages, visual representations. These representations should be easily understandable by users.
- **Handling noisy or incomplete data –**
- The data cleaning methods are required that can handle noise, incomplete objects while mining data regularities. If data cleaning methods are not there then accuracy of discovered patterns will be poor.
- **Pattern evaluation –**
- It refers to interestingness of problem. The patterns discovered should be interesting because either they represent common knowledge or lack of novelty.
- **Efficiency and scalability of data mining algorithms –**
- In order to effectively extract information from huge amount of data in databases, data mining algorithm must be efficient and scalable.
- **Parallel, distributed, and incremental mining algorithms –**
- The factors such as huge size of databases, wide distribution of data, and complexity of data mining methods motivate development of parallel and distributed data mining algorithms. These algorithms divide data into partitions that are further processed parallel. Then results from partitions are merged. The incremental algorithms update databases without having mined data again from scratch.



Data Tools

- [OpenRefine](#) – a data cleaning software that allows you to pre-process your data for analysis.
- [WolframAlpha](#) – provides detailed responses to technical searches and does very complex calculations. For business users, it presents information charts and graphs, and is excellent for high level pricing history, commodity information, and topic overviews.
- [Import.io](#) is allows you to turn the unstructured data displayed on web pages into structured tables of data that can be accessed over an API.
- [Trifacta](#) – clean and wrangle data of files & databases you could not handle in excel, with easy to use statistical tools
- [Tableau](#) – a visualization tool that makes it easy to look at your data in new ways.
- [Google Fusion Tables](#) – a versatile tool for data analysis, large data set visualization and mapping.
- [Blockspring](#) – get live data, create interactive maps, get street view images, run image recognition, and save to Dropbox with this Google Sheets plugin
- [Plot.ly](#) – visualize your data in an easy way to quickly see trends and insights
- [Luminoso](#) – identify the relationships between keywords and concepts within your data set and glean insight about product perception
- [BigML](#) – Build a model of your market, with all the variables like pricing, product features and geography



So What is a Data Scientist?



Data Scientist.....

- Summary Report for: 15-2051.00 - Data Scientists
- Develop and implement a set of techniques or analytics applications to transform raw data into meaningful information using data-oriented programming languages and visualization software. Apply data mining, data modeling, natural language processing, and machine learning to extract and analyze information from large structured and unstructured datasets. Visualize, interpret, and report data findings. May create dynamic data reports.
- Also see: Business Intelligence Analysts, Clinical Data Managers
- Tasks
 - Analyze, manipulate, or process large sets of data using statistical software.
 - Apply feature selection algorithms to models predicting outcomes of interest, such as sales, attrition, and healthcare use.
 - Apply sampling techniques to determine groups to be surveyed or use complete enumeration methods.
 - Clean and manipulate raw data using statistical software.
 - Compare models using statistical performance metrics, such as loss functions or proportion of explained variance.
 - Create graphs, charts, or other visualizations to convey the results of data analysis using specialized software.
 - Deliver oral or written presentations of the results of mathematical modeling and data analysis to management or other end users.
 - Design surveys, opinion polls, or other instruments to collect data.
 - Identify business problems or management objectives that can be addressed through data analysis.
 - Identify relationships and trends or any factors that could affect the results of research.
 - Identify solutions to business problems, such as budgeting, staffing, and marketing decisions, using the results of data analysis.
 - Propose solutions in engineering, the sciences, and other fields using mathematical theories and techniques.
 - Read scientific articles, conference papers, or other sources of research to identify emerging analytic trends and technologies.
 - Recommend data-driven solutions to key stakeholders.
 - Test, validate, and reformulate models to ensure accurate prediction of outcomes of interest.
 - Write new functions or applications in programming languages to conduct analyses.



Data Scientist.....

- Detailed Work Activities
 - Advise others on analytical techniques.
 - Analyze business or financial data.
 - Analyze data to identify or resolve operational problems.
 - Analyze data to identify trends or relationships among variables.
 - Analyze data to inform operational decisions or activities.
 - Apply mathematical principles or statistical approaches to solve problems in scientific or applied fields.
 - Determine appropriate methods for data analysis.
 - Develop procedures to evaluate organizational activities.
 - Develop scientific or mathematical models.
 - Prepare analytical reports.
 - Prepare data for analysis.
 - Prepare graphics or other visual representations of information.
 - Present research results to others.
 - Select resources needed to accomplish tasks.
 - Update technical knowledge.
 - Write computer programming code.



Data Scientist.....

- Job Zone
- Title Job Zone Four: Considerable Preparation Needed
- Education Most of these occupations require a four-year bachelor's degree, but some do not.
- Related Experience A considerable amount of work-related skill, knowledge, or experience is needed for these occupations. For example, an accountant must complete four years of college and work for several years in accounting to be considered qualified.
- Job Training Employees in these occupations usually need several years of work-related experience, on-the-job training, and/or vocational training.
- Job Zone Examples Many of these occupations involve coordinating, supervising, managing, or training others. Examples include real estate brokers, sales managers, database administrators, graphic designers, chemists, art directors, and cost estimators.



Data Scientist.....

- From US Bureau of Labor Statistics
- Industries with the highest levels of employment in Data Scientists and Mathematical Science Occupations, All Other:

Industry	Employment	% of Industry Employment	Hourly Mean Wage	Annual Mean Wage
Computer Systems Design and Related Services	9,100	.41	\$53.60	\$111,490
Management of Companies & Enterprises	7,280	.30	\$51.91	\$107,980
Management, Scientific, and technical Consulting Services	6,500	.43	\$48.46	\$100,800
Scientific Research and Development Services	2,960	.39	\$52.70	\$109,610
Insurance Carriers	2,700	.22	\$50.32	\$104,670

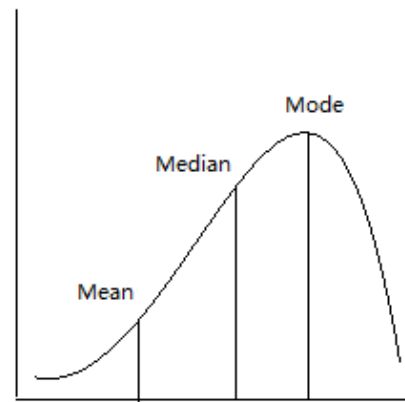


Statistics – A quick Review / Overview

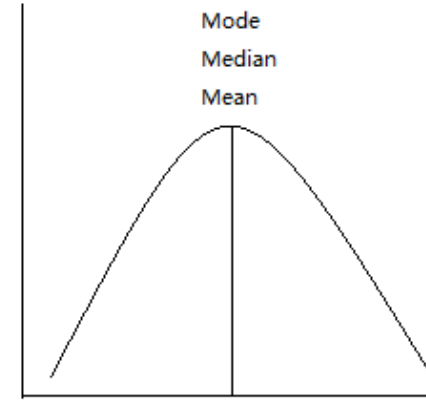
- Just a review
- It will help to understand some of the tools of data mining
 - The nice thing is that the machines will do all of the work.
 - We have to understand where to apply them.

Statistical Methods

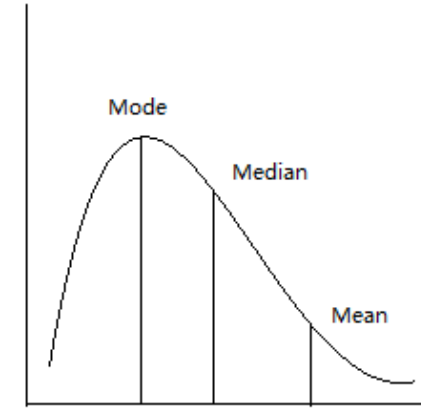
- So what are we generally using.....



Left skew



Normal Distribution



Right skew

- But don't forget about

- Mean

- The first measure we will study is the mean also known as average. Mean can be calculated by adding all data points and dividing by the number of data points.

- Median

- Median is the middle value of a sorted data set; found by ordering all data points and picking out the one in the middle (or if there are two middle numbers, taking the mean of those two numbers)

- Mode

- Mode is the most frequent number — that is, the number that occurs the highest number of times.



Statistical Methods in Data Mining

- [Geeksforgeeks](#)

Data mining refers to extracting or mining knowledge from large amounts of data. In other words, data mining is the science, art, and technology of discovering large and complex bodies of data in order to discover useful patterns. Theoreticians and

practitioners are continually seeking improved techniques to make the process more efficient, cost-effective, and accurate. Any situation can be analyzed in two ways in datamining:

- **Statistical Analysis:** In statistics, data is collected, analyzed, explored, and presented to identify patterns and trends. Alternatively, it is referred to as quantitative analysis.
- **Non-statistical Analysis:** This analysis provides generalized information and includes sound, still images, and moving images.

In statistics, there are two main categories:

- **Descriptive Statistics:** The purpose of descriptive statistics is to organize data and identify the main characteristics of that data. Graphs or numbers summarize the data. Average, Mode, SD(Standard Deviation), and Correlation are some of the commonly used descriptive statistical methods.
- **Inferential Statistics:** The process of drawing conclusions based on probability theory and generalizing the data. By analyzing sample statistics, you can infer parameters about populations and make models of relationships within data.



There are various statistical terms that one should be aware of while dealing with statistics. Some of these are:

- Quantitative Variable
- Qualitative Variable
- Discrete Variable
- Continuous Variable

Now, let's start discussing statistical methods.

This is the analysis of raw data using mathematical formulas, models, and techniques. Through the use of statistical methods, information is extracted from research data, and different ways are available to judge the robustness of research outputs.

As a matter of fact, today's statistical methods used in the data mining field typically are derived from the vast statistical toolkit developed to answer problems arising in other fields. These techniques are taught in science curriculums. It is necessary to check and test several hypotheses. The hypotheses described above help us assess the validity of our data mining endeavor when attempting to infer any inferences from the data under study. When using more complex and sophisticated statistical estimators and tests, these issues become more pronounced.

For extracting knowledge from databases containing different types of observations, a variety of statistical methods are available in Data Mining and some of these are:

- Logistic regression analysis
- Correlation analysis
- Regression analysis
- Discriminate analysis
- Linear discriminant analysis (LDA)
- Classification
- Clustering
- Outlier detection
- Classification and regression trees,
- Correspondence analysis
- Nonparametric regression,
- Statistical pattern recognition,
- Categorical data analysis,
- Time-series methods for trends and periodicity
- Artificial neural networks



Linear Regression:

The linear regression method uses the best linear relationship between the independent and dependent variables to predict the target variable. In order to achieve the best fit, make sure that all the distances between the shape and the actual observations at each point are as small as possible. A good fit can be determined by determining that no other position would produce fewer errors given the shape chosen. Simple linear regression and multiple linear regression are the two major types of linear regression. By fitting a linear relationship to the independent variable, the simple linear regression predicts the dependent variable.

Using multiple independent variables, multiple linear regression fits the best linear relationship with the dependent variable.

Classification

This is a method of data mining in which a collection of data is categorized so that a greater degree of accuracy can be predicted and analyzed. An effective way to analyze very large datasets is to classify them. Classification is one of several methods aimed at improving the efficiency of the analysis process. A Logistic Regression and a Discriminant Analysis stand out as two major classification techniques.

- **Logistic Regression:** It can also be applied to machine learning applications and predictive analytics. In this approach, the dependent variable is either binary (binary regression) or multinomial (multinomial regression): either one of the two or a set of one, two, three, or four options. With a logistic regression equation, one can estimate probabilities regarding the relationship between the independent variable and the dependent variable. For understanding logistic regression analysis in detail, you can refer to logistic regression.
- **Discriminant Analysis:** A Discriminant Analysis is a statistical method of analyzing data based on the measurements of categories or clusters and categorizing new observations into one or more populations that were identified a priori. The discriminant analysis models each response class independently then uses Bayes's theorem to flip these projections around to estimate the likelihood of each response category given the value of X. These models can be either linear or quadratic.

Discriminant Analysis

- **Linear Discriminant Analysis:** According to **Linear Discriminant Analysis**, each observation is assigned a discriminant score to classify it into a response variable class. By combining the independent variables in a linear fashion, these scores can be obtained. Based on this model, observations are drawn from a Gaussian distribution, and the predictor variables are correlated across all k levels of the response variable, Y .
- **Quadratic Discriminant Analysis:** An alternative approach is provided by Quadratic Discriminant Analysis. LDA and QDA both assume Gaussian distributions for the observations of the Y classes. Unlike LDA, QDA considers each class to have its own covariance matrix. As a result, the predictor variables have different variances across the k levels in Y .



Correlation Analysis

- In statistical terms, correlation analysis captures the relationship between variables in a pair. The value of such variables is usually stored in a column or rows of a database table and represents a property of an object.



Regression Analysis

- Based on a set of numeric data, regression is a data mining method that predicts a range of numerical values (also known as continuous values). You could, for instance, use regression to predict the cost of goods and services based on other variables. A regression model is used across numerous industries for forecasting financial data, modeling environmental conditions, and analyzing trends.

- The first step in creating good statistics is having good data that was derived with an aim in mind. There are two main types of data: an input (independent or predictor) variable, which we control or are able to measure, and an output (dependent or response) variable which is observed. A few will be quantitative measurements, but others may be qualitative or categorical variables (called factors).

Clustering

- Clustering is the process of making a group of abstract objects into classes of similar objects.
- **Points to Remember**
 - A cluster of data objects can be treated as one group.
 - While doing cluster analysis, we first partition the set of data into groups based on data similarity and then assign the labels to the groups.
 - The main advantage of clustering over classification is that, it is adaptable to changes and helps single out useful features that distinguish different groups.
- **Applications of Cluster Analysis**
 - Clustering analysis is broadly used in many applications such as market research, pattern recognition, data analysis, and image processing.
 - Clustering can also help marketers discover distinct groups in their customer base. And they can characterize their customer groups based on the purchasing patterns.
 - In the field of biology, it can be used to derive plant and animal taxonomies, categorize genes with similar functionalities and gain insight into structures inherent to populations.
 - Clustering also helps in identification of areas of similar land use in an earth observation database. It also helps in the identification of groups of houses in a city according to house type, value, and geographic location.
 - Clustering also helps in classifying documents on the web for information discovery.
 - Clustering is also used in outlier detection applications such as detection of credit card fraud.
- As a data mining function, cluster analysis serves as a tool to gain insight into the distribution of data to observe characteristics of each cluster.

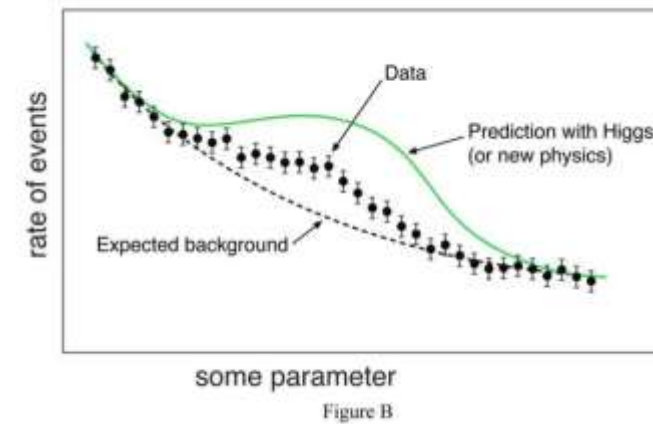
Clustering Methods

- Clustering methods can be classified into the following categories –
- Partitioning Method
 - Suppose we are given a database of ‘n’ objects and the partitioning method constructs ‘k’ partition of data. Each partition will represent a cluster and $k \leq n$. It means that it will classify the data into k groups, which satisfy the following requirements –
 - Each group contains at least one object.
 - Each object must belong to exactly one group.
- Hierarchical Method
- This method creates a hierarchical decomposition of the given set of data objects. We can classify hierarchical methods on the basis of how the hierarchical decomposition is formed. There are two approaches here –
 - Agglomerative Approach - bottom-up approach. In this, we start with each object forming a separate group. - It keeps on merging the objects or groups that are close to one another. It keep on doing so until all of the groups are merged into one or until the termination condition holds.
 - Divisive Approach - top-down approach. In this, we start with all of the objects in the same cluster. In the continuous iteration, a cluster is split up into smaller clusters.

Clustering Methods

- **Density-based Method**
 - This method is based on the notion of density. The basic idea is to continue growing the given cluster as long as the density in the neighborhood exceeds some threshold, i.e., for each data point within a given cluster, the radius of a given cluster has to contain at least a minimum number of points.
- **Grid-Based Method**
 - In this, the objects together form a grid. The object space is quantized into finite number of cells that form a grid structure.
 - Advantages
 - The major advantage of this method is fast processing time.
 - It is dependent only on the number of cells in each dimension in the quantized space.
- **Model-Based Method**
 - In this method, a model is hypothesized for each cluster to find the best fit of data for a given model. This method locates the clusters by clustering the density function. It reflects spatial distribution of the data points.
 - This method also provides a way to automatically determine the number of clusters based on standard statistics, taking outlier or noise into account. It therefore yields robust clustering methods
- **Constraint-based Method**
 - In this method, the clustering is performed by the incorporation of user or application-oriented constraints. A constraint refers to the user expectation or the properties of desired clustering results. Constraints provide us with an interactive way of communication with the clustering process. Constraints can be specified by the user or the application requirement.

Outlier Detection



- Outliers are extreme values that deviate from other observations on data , they may indicate a variability in a measurement, experimental errors or a novelty. In other words, an outlier is an observation that diverges from an overall pattern on a sample.
- Types of outliers
- Outliers can be of two kinds: univariate and multivariate. Univariate outliers can be found when looking at a distribution of values in a single feature space. Multivariate outliers can be found in a n-dimensional space (of n-features). Looking at distributions in n-dimensional spaces can be very difficult for the human brain, that is why we need to train a model to do it for us.
- Outliers can also come in different flavors, depending on the environment: point outliers, contextual outliers, or collective outliers. Point outliers are single data points that lay far from the rest of the distribution. Contextual outliers can be noise in data, such as punctuation symbols when realizing text analysis or background noise signal when doing speech recognition. Collective outliers can be subsets of novelties in data such as a signal that may indicate the discovery of new phenomena (As in figure B).

Outlier Detection – Most Common Causes

Most common causes of outliers on a data set:

- Data entry errors (human errors)
- Measurement errors (instrument errors)
- Experimental errors (data extraction or experiment planning/executing errors)
- Intentional (dummy outliers made to test detection methods)
- Data processing errors (data manipulation or data set unintended mutations)
- Sampling errors (extracting or mixing data from wrong or various sources)
- Natural (not an error, novelties in data)

In the process of producing, collecting, processing and analyzing data, outliers can come from many sources and hide in many dimensions. Those that are not a product of an error are called novelties.

Correspondence Analysis

- Correspondence analysis (CA) is an approach to representing categorical data in an Euclidean space, suitable for visual analysis. CA is often used where the data (in the form of a two-way contingency table) have many rows and/or columns and are not easy to interpret by visual inspection. The map obtained by CA helps reveal relationships within the table. CA is used in ecology, market research, and other areas.
- The initial data for CA are usually in a form of a two-way contingency table .
- The outcome of CA is numerical scores ascribed to values of the two variables - i.e to rows and columns. It is a common practice to derive two (or a greater number) of such sets of scores - each one representing a new dimension, orthogonal (zero-correlated) to the previous dimensions.



Non Parametric Regression

- Nonparametric regression is a category of regression analysis in which the predictor does not take a predetermined form but is constructed according to information derived from the data. That is, no parametric form is assumed for the relationship between predictors and dependent variable. Nonparametric regression requires larger sample sizes than regression based on parametric models because the data must supply the model structure as well as the model estimates.

Statistical Pattern Recognition

- Pattern recognition is the automated recognition of patterns and regularities in data. It has applications in statistical data analysis, signal processing, image analysis, information retrieval, bioinformatics, data compression, computer graphics and machine learning. Pattern recognition has its origins in statistics and engineering; some modern approaches to pattern recognition include the use of machine learning, due to the increased availability of big data and a new abundance of processing power. These activities can be viewed as two facets of the same field of application, and they have undergone substantial development over the past few decades.
- Pattern recognition systems are commonly trained from labeled "training" data. When no labeled data are available, other algorithms can be used to discover previously unknown patterns. KDD and data mining have a larger focus on unsupervised methods and stronger connection to business use. Pattern recognition focuses more on the signal and also takes acquisition and Signal Processing into consideration. It originated in engineering, and the term is popular in the context of computer vision: a leading computer vision conference is named Conference on Computer Vision and Pattern Recognition.

Categorical Data Analysis

- Categorical data analysis is the analysis of data where the response variable has been grouped into a set of mutually exclusive ordered (such as age group) or unordered (such as eye color) categories.
- Categorical data is data that classifies an observation as belonging to one or more categories. For example, an item might be judged as good or bad, or a response to a survey might includes categories such as agree, disagree, or no opinion.



Time Series Analysis

- Time series analysis is a specific way of analyzing a sequence of data points collected over an interval of time. In time series analysis, analysts record data points at consistent intervals over a set period of time rather than just recording the data points intermittently or randomly. However, this type of analysis is not merely the act of collecting data over time. What sets time series data apart from other data is that the analysis can show how variables change over time. In other words, time is a crucial variable because it shows how the data adjusts over the course of the data points as well as the final results. It provides an additional source of information and a set order of dependencies between the data. Time series analysis typically requires a large number of data points to ensure consistency and reliability. An extensive data set ensures you have a representative sample size and that analysis can cut through noisy data. It also ensures that any trends or patterns discovered are not outliers and can account for seasonal variance. Additionally, time series data can be used for forecasting—predicting future data based on historical data.



Neural Networks

- Neural networks are a set of algorithms, modeled loosely after the human brain, that are designed to recognize patterns. They interpret sensory data through a kind of machine perception, labeling or clustering raw input. The patterns they recognize are numerical, contained in vectors, into which all real-world data, be it images, sound, text or time series, must be translated.
- Neural networks help us cluster and classify. You can think of them as a clustering and classification layer on top of the data you store and manage. They help to group unlabeled data according to similarities among the example inputs, and they classify data when they have a labeled dataset to train on. (Neural networks can also extract features that are fed to other algorithms for clustering and classification; so you can think of deep neural networks as components of larger machine-learning applications involving algorithms for reinforcement learning, classification and regression.)
- What kind of problems does deep learning solve, and more importantly, can it solve yours? To know the answer, you need to ask a few questions:
 - What outcomes do I care about? In a classification problem, those outcomes are labels that could be applied to data: for example, spam or not_spam in an email filter, good_guy or bad_guy in fraud detection, angry_customer or happy_customer in customer relationship management. Other types of problems include anomaly detection (useful in fraud detection and predictive maintenance of manufacturing equipment), and clustering, which is useful in recommendation systems that surface similarities.
 - Do I have the right data? For example, if you have a classification problem, you'll need labeled data. Is the dataset you need publicly available, or can you create it (with a data annotation service like Scale or AWS Mechanical Turk)? In this example, spam emails would be labeled as spam, and the labels would enable the algorithm to map from inputs to the classifications you care about. You can't know that you have the right data until you get your hands on it. If you are a data scientist working on a problem, you can't trust anyone to tell you whether the data is good enough. Only direct exploration of the data will answer this question.

Conditional Probability

- If you toss a coin, do the previous events have any effect on the next flip?
 - If you toss it and get heads 3 times in a row – what is the odds that it will be heads on the next throw
 - 50%
- What are the odds of 4 in a row - $.5 * .5 * .5 * .5 = 6.25\%$
 - But that is from the beginning of the series – we are measuring not one event but four



Conditional Probability

- Conditional probability could describe an event like:
- Event A is that it is raining outside, and it has a 0.3 (30%) chance of raining today.
- Event B is that you will need to go outside, and that has a probability of 0.5 (50%).
- A conditional probability would look at these two events in relationship with one another, such as the probability that it is both raining and you will need to go outside.
- The formula for conditional probability is:
- $P(B|A) = P(A \text{ and } B) / P(A)$
- which you can also rewrite as:
- $P(B|A) = P(A \cap B) / P(A)$



Conditional Probability - Examples

- Imagine that you're a furniture salesman. The probability of a new customer to your store purchasing a couch on any particular day is 30%. However, if they are entering your store in the month leading up to the Super Bowl, the probability might be 70%. We can represent the conditional probability of selling a couch if it is the month leading up to Super Bowl as $P(\text{Selling a couch} \mid \text{Super Bowl month})$ where the $\mid$ symbol means "given that". This conditional probability gives us a way to express probabilities when our beliefs change about the probability of one event happens (a couch sale in this example) given that a certain event has happened (in this case, the advent of the month preceding the Super Bowl).
- The probability of a woman between 40 and 50 years of having breast cancer is about 1%. However, this probability changes if a woman has a positive mammogram: the probability a woman has cancer if she has a positive mammogram result rises to about 8.3%.
- Four candidates A, B, C, and D are running for a political office. Each has an equal chance of winning: 25%. However, if candidate A drops out of the race due to ill health, the probability will change: $P(\text{Win} \mid \text{One candidate drops out}) = 33.33\%$.
- If you're pregnant, the probability of having a boy or girl is the same: 50%. However, if you already have one child (say, a boy), your odds change. Given that your first child is a boy, your odds of having another boy drop to one third (33.33%). The reason for this is that the sample space for the event you have one boy out of two is $S = \{BB, BG, GB\}$. If you have one boy, the only possible event within this space is BB, which is one third of the sample space.
 - That assumes that boys and girls are even probabilities – are there other factors?
 - Anatomy – why do some families have significantly more of one sex
 - Statistics work over large populations



Bayes Theorem

- Bayes' theorem is a way to figure out conditional probability. Conditional probability is the probability of an event happening, given that it has some relationship to one or more other events. For example, your probability of getting a parking space is connected to the time of day you park, where you park, and what conventions are going on at any time. Bayes' theorem is slightly more nuanced. In a nutshell, it gives you the actual probability of an event given information about tests.
- "Events" Are different from "tests." For example, there is a test for liver disease, but that's separate from the event of actually having liver disease.
- Tests are flawed: just because you have a positive test does not mean you actually have the disease. Many tests have a high false positive rate. Rare events tend to have higher false positive rates than more common events. We're not just talking about medical tests here. For example, spam filtering can have high false positive rates. Bayes' theorem takes the test results and calculates your real probability that the test has identified the event.

- **The Formula**

- Bayes' Theorem (also known as Bayes' rule) is a deceptively simple formula used to calculate conditional probability. The Theorem was named after English mathematician Thomas Bayes (1701-1761). The formal definition for the rule is:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- bayes' theorem
- In most cases, you can't just plug numbers into an equation; You have to figure out what your "tests" and "events" are first. For two events, A and B, Bayes' theorem allows you to figure out $p(A|B)$ (the probability that event A happened, given that test B was positive) from $p(B|A)$ (the probability that test B happened, given that event A happened). It can be a little tricky to wrap your head around as technically you're working backwards; you may have to switch your tests and events around, which can get confusing. An example should clarify what I mean by "switch the tests and events around."



Bayes Theorem

- Bayes' Theorem Example #1
- You might be interested in finding out a patient's probability of having liver disease if they are an alcoholic. "Being an alcoholic" is the test (kind of like a litmus test) for liver disease.
- A could mean the event "Patient has liver disease." Past data tells you that 10% of patients entering your clinic have liver disease. $P(A) = 0.10$.
- B could mean the litmus test that "Patient is an alcoholic." Five percent of the clinic's patients are alcoholics. $P(B) = 0.05$.
- You might also know that among those patients diagnosed with liver disease, 7% are alcoholics. This is your $B|A$: the probability that a patient is alcoholic, given that they have liver disease, is 7%.
- Bayes' theorem tells you:
- $P(A|B) = (0.07 * 0.1)/0.05 = 0.14$
- In other words, if the patient is an alcoholic, their chances of having liver disease is 0.14 (14%). This is a large increase from the 10% suggested by past data. But it's still unlikely that any particular patient has liver disease.

Clustering – K Means

- K-means is an algorithm that trains a model that groups similar objects together. The k-means algorithm accomplishes this by mapping each observation in the input dataset to a point in the n-dimensional space (where n is the number of attributes of the observation). For example, your dataset might contain observations of temperature and humidity in a particular location, which are mapped to points (t, h) in 2-dimensional space.
- In k-means clustering, each cluster has a center. During model training, the k-means algorithm uses the distance of the point that corresponds to each observation in the dataset to the cluster centers as the basis for clustering. You choose the number of clusters (k) to create.
- For example, suppose that you want to create a model to recognize handwritten digits and you choose the MNIST dataset for training. The dataset provides thousands of images of handwritten digits (0 through 9). In this example, you might choose to create 10 clusters, one for each digit (0, 1, ..., 9). As part of model training, the k-means algorithm groups the input images into 10 clusters.



Next week

- Discussion questions for this week
 - Can you find a prediction error or blunder? Describe it and the consequences
 - Can you find an example of a successful data mining project?
 - How do you define it as successful?
- Next Week
 - Exploratory Statistics